Lecture 25: Three-phase Systems - Examples

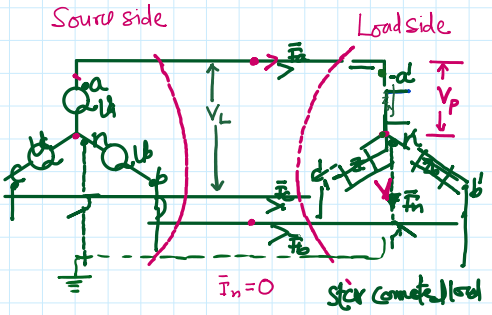
Example1: Consider a Y- connected load $Z_a = Z_b = Z_c = 10\angle 45^\circ\ \Omega = 7+j7\ \Omega$

$\bar{V}_a = 100\angle 0^\circ\ V$

Find out powers phase wise, pf, and 3φ values

$$\bar{I}_a = \frac{\bar{V}_a}{Z_a} = \frac{100\angle 0^\circ}{10\angle 45^\circ}$$

$$\boxed{\bar{I}_a = 10\angle -45^\circ\ A}$$



$$\bar{S} = 3 \times V \times (I\angle\phi)^*$$

$$= 3 \times 100 \times 10\ \angle 45^\circ$$

$$= 3000\angle 45^\circ = 2121 + j\,2121$$

$$= P + jQ$$

$$P_{3\phi} = P = 2121\ W$$

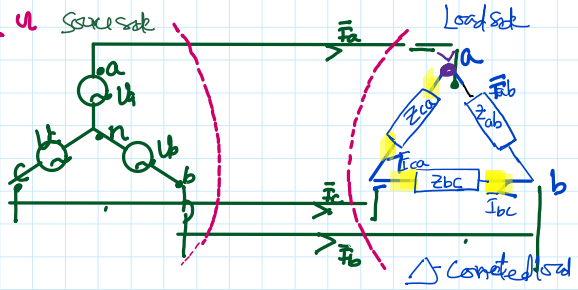
$$Q_{3\phi} = Q = 2121\ VArs$$

$$pf = \cos(45^\circ)\ (lag) = 0.707\ (lag)$$

Example2: Now Z_a, Z_b, Z_c are connected in Δ- configuration

Compute: P, Q, S, pf, Line to Line parameters V_L, I_L?

$Z_{ab} = Z_{bc} = Z_{ca} = Z = 10\angle 45^\circ\ \Omega$



$$\bar{V}_{ab} = \sqrt{3}V\angle 30^\circ$$

$$\bar{I}_{ab} = \frac{\bar{V}_{ab}}{Z_{ab}} = \frac{\sqrt{3}V\angle 30^\circ}{10\angle 45^\circ}$$

$$\bar{I}_{ab} = 10\sqrt{3}\angle -15^\circ$$

$$\bar{S} = 3\,\bar{V}_{ab}\,\bar{I}_{ab}^* = 3 * \sqrt{3}V\angle 30^\circ * (10\sqrt{3}\angle -15^\circ)^*$$

$$= 3 \times 3 \times V \times I^*\ \angle 45^\circ$$

$$= 9000\angle 45^\circ$$

$$\bar{S} = 6363 + j\,6363$$

$$\Rightarrow P_{3\phi} = 6363\ W,\ Q_{3\phi} = 6363\ VArs$$

$\Rightarrow P_{3\phi} = 6363\ W,\ Q_{3\phi} = 6363\ VARs$

$pf = \cos 45^\circ$ (lag)

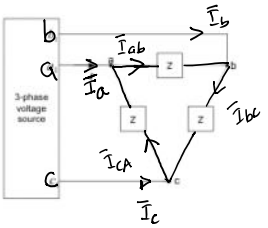
$= 0.707$ (lag)

Q9. A 220-V (L-L), three phase balanced voltage is applied to a balanced delta-connected three phase load in the manner illustrated in the figure below. The phase current measured between points *a* and *b* is $\bar{I}_{ab} = 10\angle -30^0$ A with $\bar{V}_{ab} = 220\angle 0^0$ V as reference phasor.

(a) Find the phase and line currents. Draw the phasor diagram showing clearly the line voltages, phase currents and line currents.

(b) Compute the total power received by the three-phase load.

(c) Find the value of the resistive portion of the phase impedance.



[Ans.: (a) $10\angle -30^0$ A, $10\angle -150^0$ A, $10\angle 90^0$ A, $17.32\angle -60°$ A, $17.32\angle -180°$ A, $17.32\angle -300°$ A (b) 5715.6 W, 2857.8 Var (lag) (c) 19.05 Ω

Solution: Phase currents $\bar{I}_a = 10\angle -30^\circ$ A

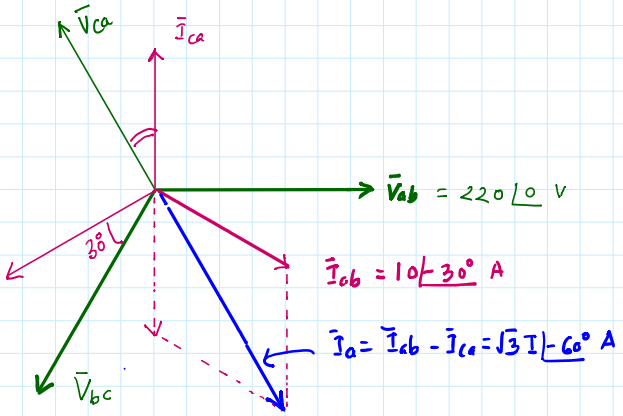
$\bar{I}_b = 10\angle -150^\circ$ A

$\bar{I}_c = 10\angle 90^\circ$ A

Line currents: $\bar{I}_a = \bar{I}_{ab} - \bar{I}_{ca} = \sqrt{3}\times 10\angle -60^\circ = 17.32\angle -60^\circ$

$\bar{I}_b = \bar{I}_{bc} - \bar{I}_{ab} = 17.32\angle -180^\circ$ A $= -17.32\angle 0$

$\bar{I}_c = \bar{I}_{ca} - \bar{I}_{bc} = 17.32\angle -300^\circ = 17.32\angle 60^\circ$ A



(b) Powers: $\bar{S} = \bar{V}_{ab}\bar{I}_{ab}^* + \bar{V}_{bc}\bar{I}_{bc}^* + \bar{V}_{ca}\bar{I}_{ca}^*$

$= 3\,\bar{V}_{ab}\bar{I}_{ab}^* = 3\times 220\times(10\angle -30)^* = 6600\angle 30^\circ$

$= 5715.76 + j\,3300$

$P = 5715.76\,W$, $Q = 3300\,VAR$

$S = 6600\,VA$, $pf = P/S = \cos 30^\circ = 0.866$ (lag.)

(c) $\bar{Z}_{ab} = \dfrac{\bar{V}_{ab}}{\bar{I}_{ab}} = \dfrac{220\angle 0}{10\angle -30^\circ} = 22\angle 30^\circ = 19.05 + j\,11$

$R_{ab} = 19.05\,\Omega$

$X_{ab} = 11\,\Omega$ (inductive)

Q10. The delta-connected load of Fig. 1 consists of phase impedance each equal to $15 + j20\ \Omega$.

(a) Find the phasor current in each line.
(b) What is the power consumed per phase?
(c) What is the phasor sum of the three phase line currents? Why does it have this value?

[Ans.: (a) $8.8\angle -53.13^\circ$, $8.8\angle -173.13^\circ$, $8.8\angle 66.86^\circ$ A (b) 1161.6 W, 1548.8 Var (lag) (c) 0 A

Solution: Similar to Q9, except $\bar{Z}_{ab} = \bar{Z}_{bc} = \bar{Z}_{ca} = 15 + j20\,\Omega$

$= 25\angle 53.13^\circ\,\Omega$

So $\bar{I}_{ab} = \dfrac{\bar{V}_{ab}}{\bar{Z}_{ab}} = \dfrac{220\angle 0}{25.0\angle 53.13^\circ} = 8.8\angle -53.13^\circ$

$\bar{I}_{bc} = 8.8\angle -173.13^\circ$ A

$\bar{I}_{ca} = 8.8\angle 66.87^\circ$ A

Line currents: $\bar{I}_a = 15.24\angle -83.13^\circ$ A

$\bar{I}_b = 15.24\angle 156.87^\circ$ A

$\bar{I}_c = 15.24\angle 36.87^\circ$ A

(b) Powers Consumed per phase:

$\bar{S}_{ab} = \bar{V}_{ab}\,\bar{I}_{ab}^{\,*} = 220\angle 0 \cdot (8.8\angle -53.13)^*$

$= 1161.6 + j\,1548$

$P_{ab} = 1161.6\,W = P_{bc} = P_{ca}$

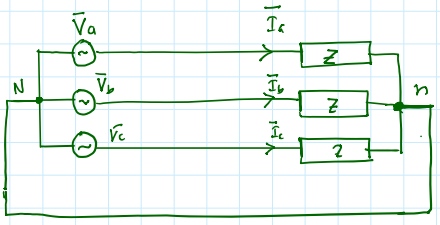
$Q_{ab} = 1548\,VAr\,(lag) = Q_{bc} = Q_{ca}$

(c) $\bar{I}_a + \bar{I}_b + \bar{I}_c = 0$ (As currents are balanced)

$\bar{I}_{ab} + \bar{I}_{bc} + \bar{I}_{ca} = 0$

Q11. A three-phase, 208 V (L-L) generator supplies a total 1800 W at a line current of 10 A. When three identical impedances are arranged in a star connection across the line terminals of the generator. Compute the resistive and reactive components of each phase impedance.

[**Ans.**: 6 Ω, ±10.39 Ω]



Solution: $V_{L\text{-}L} = 208$

$V_P = V = 208/\sqrt{3}\ V$

$\bar{I}_a = 10\angle \varnothing$ A

$P_{3\phi} = 3VI\cos\phi$

$1800 = 3 \times \frac{208}{\sqrt{3}} \times 10 \cos\phi$

$\Rightarrow \cos\phi \quad \frac{1800}{\sqrt{3} \times 208 \times 10} = 0.49 \simeq 0.5$

$\phi = \cos^{-1}(0.5) = \pm 60^\circ$ (lag or lead)

$I = |\bar{I}| = \left|\frac{\bar{V}}{\bar{Z}}\right| = \frac{V}{|\bar{Z}|} \Rightarrow \frac{208/\sqrt{3}}{|Z|} = 10$

$|Z| = 12\,\Omega$

$\therefore \quad \bar{Z} = |Z| \angle \pm 60^\circ = 12 \angle \pm 60^\circ$

$R = 6\,\Omega, \quad X = \pm 10.39\,\Omega$ per phase.

Q12. A balanced three-phase star-connected load has an impedance of $4\angle 60^\circ\ \Omega$ from line to neutral. Moreover, from line *a* to neutral *n* the voltage in $\bar{V}_{an} = 20\angle 30^\circ$ V.

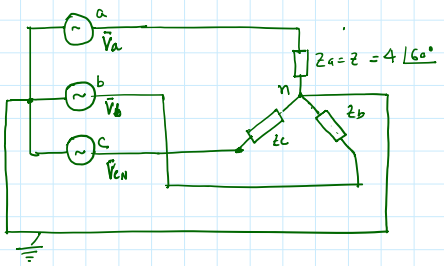
(a) What is the current in phases a, *b* and *c*?

(b) What is the voltage from line *b* to neutral *n*?

(c) What is the phasor expression for the voltage from line *a* to line *c*, i.e., $\bar{V}_{ac}$?

[Ans.: (a) $5\angle -30^\circ$, $5\angle -150^\circ$, $5\angle 90^\circ$ A (b) $20\angle -90^\circ$ V (c) $34.64\angle 0^\circ$ V]

Solution:



$\bar{I}_a = \frac{\bar{V}_{AN}}{Z_a} = \frac{20\angle 30^\circ}{4\angle 60^\circ} = 5\angle -30^\circ$ A

$\bar{I}_b = 5\angle -150^\circ$ A

$\bar{I}_c = 5\angle 90^\circ$ A

(b) $\bar{V}_{bN} = 20\angle 30^\circ - 120^\circ = 20\angle -90^\circ$

$\bar{V}_{CN} = 20\angle 150^\circ$

(c) $\bar{V}_{ac} = \bar{V}_a - \bar{V}_c = 20\angle 30^\circ - 20\angle 150^\circ$

$= 34.64\angle 0^\circ$

